



Identifying nitrate sources and nitrogen transformation in a karst river–groundwater system using dual nitrate isotopes and a Bayesian (SIAR) mixing model

Zupeng Wan¹ · Baojun Liu² · Wenwen Chen¹ · Yingjie Chen³ · Xiaoyu Yan⁴ · Honghu Zeng¹

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Abstract

Karst basins have permeable surface–subsurface connections that can accelerate nitrogen transport. The Lipu River Basin in southern Guilin is affected by intensive agriculture and industrial activity, but its nitrate sources and transformations remain poorly resolved. We integrated dual nitrate isotopes ($\delta^{15}\text{N-NO}_3^-$ and $\delta^{18}\text{O-NO}_3^-$), hydrochemical data, and a Bayesian SIAR mixing model. River-water TN was significantly higher in winter (7.09 ± 6.59 mg/L) than in summer (2.95 ± 1.48 mg/L). River-water NO_3^- -N was lower in summer than in winter and lower than groundwater only during summer; the winter river–groundwater difference was not significant. The highest individual groundwater NO_3^- -N values occurred near the industrial park, but the land-use group difference was not significant (Kruskal–Wallis, $p=0.48$). The SIAR model identified manure and sewage as the largest modeled summer nitrate source, contributing 49% at the basin scale. These results identify elevated winter river loads and groundwater nitrate hotspots near tributaries and industrial zones as management priorities.

Keywords Bayesian isotope mixing model · Nitrate source apportionment · Karst aquifer · Uncertainty quantification · Environmental risk assessment

Introduction

Pollution-control strategies developed for porous-media-dominated basins are often inadequate for heterogeneous karst environments (Guo et al. 2019). When industrial and agricultural nitrogen inputs interact with karst hydrology,

observed transport may depart from predictions based on porous-media models (P. Zhang et al. 2023). A highly permeable karst conduit network enables refractory organic matter in industrial wastewater and agricultural nitrogen to bypass the slow degradation process in the soil zone for rapid and deep transboundary transport (Güven et al. 2017). Rather than attenuating contaminants, these conduits may accelerate their transport and transformation, thereby increasing ecological exposure and potential toxicity (Wells et al. 2018).

Most nitrogen source-apportionment studies have examined non-karst catchments dominated by porous media (Jost et al. 2011; Li et al. 2022). Studies in karst systems have focused mainly on agricultural or single-land-use catchments. Where dual nitrate isotopes and Bayesian mixing models have been applied to karst and plateau systems in Southwest China, manure & sewage and soil nitrogen have repeatedly emerged as the dominant NO_3^- sources (Fan et al. 2025; Liu et al. 2025); still, basins in which intensive industry is superimposed on dense rural land use—where point- and non-point inputs are strongly mixed—remain

✉ Honghu Zeng
zenghonghu@glut.edu.cn

¹ Guangxi Key Laboratory of Environmental Pollution Control Theory and Technology, Guilin University of Technology, Guilin 541006, China

² Hubei College of Water Resources and Hydropower Technology, Wuhan 430000, China

³ SCNU Environmental Research Institute, Guangdong Provincial Key Laboratory of Chemical Pollution and Environmental Safety & MOE Key Laboratory of Theoretical Chemistry of Environment, South China Normal University, Guangzhou 510006, China

⁴ Changhu Ecological Administration Bureau, Jingzhou 434000, China

poorly characterised. Discrete conduit flow and rapid surface-subsurface exchange challenge lumped and semi-distributed watershed models such as SWAT and HEC-HMS. These models represent the subsurface as a continuous porous medium and may not capture heterogeneous flow paths or nitrogen transport reliably (Iriawan H et al. 2021). For source apportionment, dual NO_3^- isotopes are increasingly combined with Bayesian mixing models, the IsoSource mass-balance approach, or statistical receptor models such as PMF and APCS-MLR. Most applications have focused on non-karst settings. Few studies have quantified source proportions and their uncertainty in coupled karst river-groundwater systems, where conduits create heterogeneous links between surface water and groundwater. We therefore investigated the Lipu River Basin, where nitrate concentrations are elevated but source contributions remain unquantified. We combined dual nitrate isotopes with hydrochemical data to trace sources and assess transformation pathways in this karst river-groundwater system.

Dissolved inorganic nitrogen (DIN) in natural waters occurs mainly as NO_3^- , NO_2^- and NH_4^+ . Owing to its high mobility, NO_3^- is typically the dominant DIN species and a sensitive indicator of pollutant transport in karst systems (Zuo et al. 2023). In addition, high NO_3^- levels are associated with methemoglobinemia in humans and aquatic ecosystem degradation (Yan et al. 2023). NO_3^- pollution sources in karst basins include point and river sources. Point sources include mainly industrial parks and landfills. In addition, NO_3^- concentrations and sources are affected by various nitrogen transformations. For example, higher NH_4^+ concentrations can promote nitrification (Ma et al. 2021). They may also reduce net NO_3^- removal by assimilation, thereby promoting nitrate accumulation in karst basins. Fertilizer application commonly dominates nitrate inputs in agricultural areas (Wang et al. 2024), whereas point sources are often more important in urbanized areas (Lai et al. 2023). However, manure, sewage, and soil nitrogen can also dominate groundwater NO_3^- inputs in agricultural areas. Integrating multiple tracers with source-apportionment models can quantify these contributions and distinguish natural from anthropogenic controls.

NO_3^- sources largely reflect basin nitrogen sources (Iqbal et al. 2022; Schaefer et al. 2016). Linking land use types to cations and anions has been a traditional approach to identifying NO_3^- sources and nitrogen cycling processes (Biddau et al. 2023). The NO_3^- stable isotope method ($\delta^{15}\text{N}-\text{NO}_3^-$ and $\delta^{18}\text{O}-\text{NO}_3^-$) provides meaningful insights into NO_3^- sources and nitrogen conversion processes because different NO_3^- sources exhibit unique isotopic characteristics (Xue et al. 2009). The NO_3^- stable isotope method requires minimal auxiliary information, offers high accuracy and operational simplicity, and directly identifies NO_3^- sources (Yang et

al. 2024). In general, manure and sewage have relatively high $\delta^{15}\text{N}-\text{NO}_3^-$ values, whereas atmospheric deposition has high $\delta^{18}\text{O}-\text{NO}_3^-$ values. Mineral fertilizer and soil nitrogen usually have lower $\delta^{15}\text{N}-\text{NO}_3^-$ values, which helps constrain nitrate sources in aquatic systems (Wu et al. 2022). However, isotope signatures can overlap between fertilizer and soil nitrogen and between soil nitrogen and manure or sewage. Source mixing and isotope fractionation during biogeochemical transformations further limit source apportionment (Lin et al. 2019). Consequently, NO_3^- isotopes are often combined with complementary tracers. In this study, auxiliary tracers were used to distinguish mixing from denitrification within the analytical scope, while the potential value of additional tracers and integrated approaches is acknowledged in the Conclusion (Grechanik et al. 2024; Li et al. 2023).

This study integrated a Bayesian isotope mixing model with NO_3^- stable isotope analysis. Additional covariates (hydrochemical parameters and land use types) enhanced source apportionment accuracy compared with traditional mixing models. The hydrological status of the southwestern karst basin is complex, with intense agricultural and human activities. Groundwater, rainfall, and river water are key sources of river recharge. Towns and farmlands are scattered around karst basin (Chang et al. 2024). Therefore, karst basins are typical areas of mixed land use (Yi et al. 2015). Frequent seasonal groundwater-river water exchange results in severe groundwater nitrogen pollution, with average nitrogen and NO_3^- concentrations reaching 24.35 and 15.15 mg/L (Boumaiza et al. 2022), respectively, posing a major threat to lake water quality and safety. In coupled karst river-groundwater systems it remains unclear (i) how the dominant NO_3^- sources differ between river water and groundwater, (ii) whether nitrification or denitrification controls the NO_3^- pool, and (iii) what the proportional source contributions-with their uncertainty-actually are.

To address these research gaps, we investigated inorganic nitrogen pollution in river water and groundwater in the Lipu River Basin within Lipu County, considering domestic sewage, industrial sewage, and other pollution sources. By integrating NO_3^- stable isotope ($\delta^{15}\text{N}$ and $\delta^{18}\text{O}$) analysis with water chemistry data, meteorological parameters, environmental factors, and the Bayesian isotope mixing (SIAR, Stable Isotope Analysis in R) model, the objectives of this study are to: (1) clarify the distribution characteristics of DIN ($\text{NO}_3^--\text{N}+\text{NO}_2^--\text{N}+\text{NH}_4^+-\text{N}$) species under the influence of various discharge types in a typical karst area; (2) investigate the spatiotemporal distribution of nitrogen-containing pollutants in river water and groundwater and their correlations with comprehensive hydrochemical indicators; and (3) identify the sources of

NO_3^- based on nitrogen distribution characteristics and the SIAR model. The results can inform evaluation and management of nitrogen pollution across contrasting discharge settings in karst regions.

Materials and methods

Description of the study area

The study area lies in Lipu City, southern Guilin, in north-eastern Guangxi Zhuang Autonomous Region. It extends into western Pingle County and reaches the western banks of the Lijiang and Guijiang rivers. The basin spans $110^\circ 12' 6''$ – $110^\circ 40' 57''$ E and $24^\circ 20' 54''$ – $24^\circ 39' 27''$ N. The region has distinct wet and dry seasons, with long, hot, rainy summers and generally mild winters. The long-term mean temperature is 18.9°C , and mean annual precipitation is 1,441.5 mm, concentrated from April to August. The submerged aquifer in the area is 0.5–3.5 m deep. The lithology of the area is composed mainly of clastic, carbonate, igneous, and loose rocks. The Quaternary Holocene Q_4 is located on the upper terrace, with pore aquifers and partial pressure. The northern part of the basin overlies the Lower Carboniferous stages of the Datangian and Yanguandian C_1 , which contain fissure cave water with a small flow of springs (Zhang et al. 2020).

Watershed monitoring

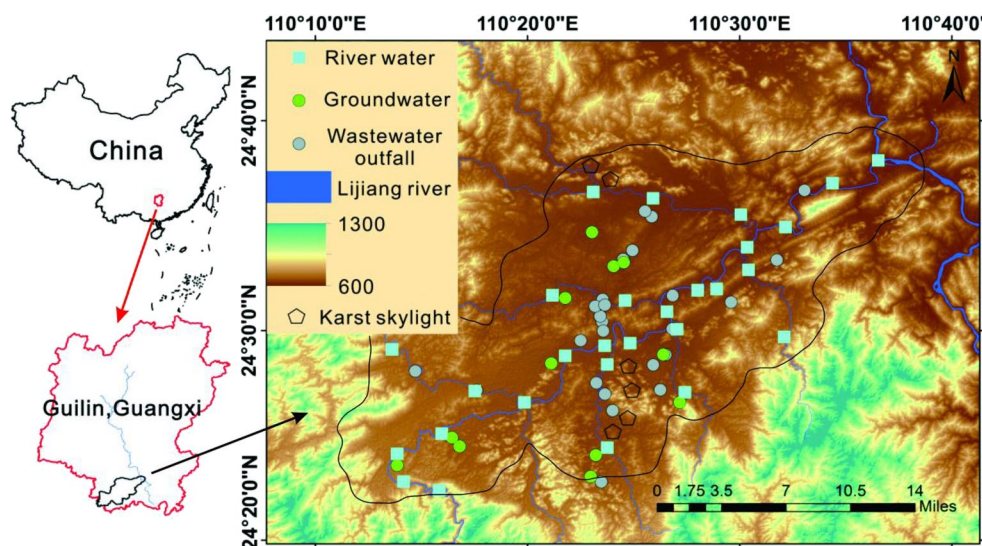
Two sampling campaigns were conducted in 2022, one during the wet season (summer) and one during the dry season (winter; Fig. 1). The summer campaign included 27 river sites and 12 groundwater sites. The winter campaign included 24 river sites and 14 groundwater sites, yielding

51 river-water and 26 groundwater samples in total. Mainstream sites were distributed from upstream Tianjingling to the downstream Guijiang confluence. Tributary sites covered the Pulu, Dumo, Datang, Xinping, Limu, and Maling rivers. Groundwater sites followed the inferred flow direction and represented three aquifer types: Quaternary porous aquifers, Carboniferous fissure-cave aquifers, and bedrock-fissure aquifers. Table S1 lists all sampling sites, coordinates, water-body types, and remarks. Text S1 describes the hydrogeological setting.

Sample collection

Sampling sites covered the basin's recharge, runoff, and discharge zones and were classified by dominant surrounding land use and discharge type (industrial, sanitary, or other). These categories were based on land use rather than direct measurements of flow or pollutant load at individual outlets. One local effluent sample (LTE) was collected at the location listed in Table S1. The summer isotope dataset comprised 39 mixture samples (27 river-water and 12 groundwater samples) plus LTE. LTE was used only to assess whether local effluent fell within the literature-defined manure-and-sewage prior; it was not entered as a mixture sample or as a separate modeled source. Before collection, brown glass bottles were rinsed three times with ultrapure water and three times with sample water, then sealed with aluminum foil and capped. Duplicate samples represented at least 10% of sites in each sampling batch. Groundwater was collected from phreatic wells and karst springs that tap the shallow aquifer. The water table was 0.5–3.5 m below ground surface. Wells were purged until pH, electrical conductivity, and temperature stabilized, after which samples were collected 0.5–1.0 m below the water table.

Fig. 1 Overview of the Lipu River Basin study area



Sample preparation and analysis

Detection of inorganic carbon and nitrogen compounds

Immediately after the samples were transported to the laboratory, $\text{NH}_4^+\text{-N}$, $\text{NO}_3^-\text{-N}$, $\text{NO}_2^-\text{-N}$, and inorganic carbon (DIC, including CO_3^{2-} and HCO_3^-) were measured. The remaining samples were filtered using a 0.45- μm filter membrane (Tianjin Jinteng, China) and stored in the dark at 0–4 °C. DIN concentrations were measured spectrophotometrically with a JH-TD401 multiparameter water-quality analyzer. DIC was measured using total-alkalinity test kits (10–200 mg/L, Lu Heng Bio, China).

Hydrochemical detection and data analysis

The anion and cation indicators to be tested included K^+ , Na^+ , Ca^{2+} , Mg^{2+} , SO_4^{2-} , and Cl^- . The samples to be tested were filtered using a 0.22- μm filter membrane and stored away from light at 4 °C. Cationic concentrations were determined using an inductively coupled plasma emission spectrometer (Optima 8000, USA) in accordance with the method specified in the National Environmental Protection Standard (HJ 776–2015, China). Anions were detected using ion chromatography (DIONEX ICS-2100, USA). Anions were determined using the method specified in the National Environmental Protection Standard (HJ 84-2016, China). Because the data were non-normally distributed, non-parametric tests were used throughout. The Friedman test compared nitrogen species measured in the same samples. Mann-Whitney U and Kruskal-Wallis tests were used for two-group and multigroup comparisons, respectively, and Spearman's rank correlation assessed associations between solutes. Significance was set at $p < 0.05$. A summary of the key tests is given in Table S5.

Sampling and analysis of dual nitrate isotopes ($\delta^{15}\text{N}\text{-NO}_3^-$ and $\delta^{18}\text{O}\text{-NO}_3^-$)

Sample collection for NO_3^- isotopes

River water samples were collected onsite using PET sample bottles. First, they were filtered through a 0.45 μm filter membrane to remove various suspended solids and microorganisms. Subsequently, the water samples were stored at temperatures below 4 °C. Dual NO_3^- isotopes were determined by the bacterial denitrifier method, in which sample NO_3^- is converted to N_2O by denitrifying bacteria and the N_2O analysed by isotope-ratio mass spectrometry. Results were calibrated against the international reference materials USGS32, USGS34 and USGS35 (with IAEA- NO_3^- as a check); the analytical precision (1σ) was better than $\pm 0.3\text{‰}$

for $\delta^{15}\text{N}$ and $\pm 0.5\text{‰}$ for $\delta^{18}\text{O}$. Text S3 summarizes method detection limits and the precision of the hydrochemical measurements. Samples were filtered through 0.22- μm cartridge filters, and nitrate concentrations were measured before the samples were frozen and stored. An aliquot of the denitrifying bacterial culture was plated and incubated at 30 °C for 3 days. A single colony was then transferred to 6 mL of TSB medium and incubated for 8 h at 32 °C and 180 rpm. The remaining strain was stored at 4 °C.

Sample pre-treatment and isotope determination

A 0.2-mL aliquot of culture was added to 10 mL of tryptic soy broth (TSB) and incubated under the same conditions for 16 h. Next, 10 mL of this culture was transferred to 238 mL of TSB and incubated for 48 h. A colorimetric test was performed after culture expansion. The next day, 0.2 mL of 10 mol/L NaOH was injected to terminate the reaction and absorb CO_2 . The generated gas was then analyzed instrumentally.

Hydrogen and oxygen isotopes were measured using a liquid water isotope laser spectrometer. Stable carbon isotope ($\delta^{13}\text{C}$) analyses were performed by the Karst Geological Resources and Environment Supervision and Testing Center, Institute of Karst Geology, Chinese Academy of Geological Sciences, Guilin.

Nitrogen Enrichment Slope (NES)

The nitrogen enrichment slope (NES) is the slope b of the regression of $\log_{10}\text{C}(\text{NO}_3^-)$ on $\delta^{15}\text{N}$ (Eq. 1; after Lavoie et al. 2013) and indicates how $\delta^{15}\text{N}$ changes as NO_3^- concentration changes; a positive slope ($\text{NES} > 0$) is consistent with progressive nitrogen accumulation/enrichment along the flow path.

$$\log C_{\text{NO}_3^-} = \delta^{15}\text{N}(b) + a \quad (1)$$

where the slope (b) represents the NES. A positive slope ($\text{NES} > 0$) indicates increased nitrogen accumulation.

SIAR configuration, priors and uncertainty

A Bayesian stable isotope mixing model was used to quantify the contributions of different nitrate sources to the Lipu River Basin water system. The model was implemented with the Stable Isotope Analysis in R (SIAR) package and was formulated as follows (Ji et al. 2022):

$$X_{ij} = \sum p_k (S_{ij} + C_{ij}) + \varepsilon_{ij} \quad (2)$$

$$S_{ij} = N(\mu_{ij}, \omega_{jk}^2) \quad (3)$$

$$C_{ij}N(\lambda'_{ij}, \omega\tau'^2_{jk}) \quad (4)$$

$$\varepsilon_{ij} = N(0, \sigma_j^2) \quad (5)$$

In the above formula (Eqs. 1, 2, 3, 4, 5), X_{ij} is the j -th isotope value of the i -th mixture, $i=1,2,3,\dots,N$; $j=1,2,3,\dots,J$; P_k is the proportion of the k -th source that needs to be estimated using the SIAR model; S_{ij} is the value of the j -th isotope from the i -th source, following a normal distribution with a mean of μ and a variance of ω ; C_{ij} is the fractionation coefficient of the j -th isotope from the i -th source, following a normal distribution with a mean of λ and a variance of τ ; ε_{ij} is the residual error, representing the variance that cannot be quantified among the other mixtures, with the mean and variance typically being 0.

To estimate the proportional contributions of NO_3^- -N sources, two isotopes ($j=2$; $\delta^{15}\text{N}-\text{NO}_3^-$ and $\delta^{18}\text{O}-\text{NO}_3^-$) and three potential end-members were considered: nitrogen fertilizer (NF), soil nitrogen (SN), and manure & sewage (M&S) (She et al. 2024; Liu et al. 2025). The $\delta^{15}\text{N}-\text{NO}_3^-$ and $\delta^{18}\text{O}-\text{NO}_3^-$ signatures (mean $\pm$ SD) assigned to each end-member were compiled from previously published values for comparable settings and are summarised in Table S6.

Source $\delta^{15}\text{N}-\text{NO}_3^-$ and $\delta^{18}\text{O}-\text{NO}_3^-$ prior means and standard deviations were taken from widely cited literature compilations (Table S4) and supplemented by a local effluent end-member (LTE) measured in this study, one local effluent end-member (LTE) was collected directly from a karst basin effluent discharge point (Table S1; coordinates 110.482°E, 24.533°N). The measured $\delta^{15}\text{N}-\text{NO}_3^-$ (+8.6‰) and $\delta^{18}\text{O}-\text{NO}_3^-$ (−1.3‰) of LTE fall within the manure-and-sewage (M&S) literature range (Table S4), confirming its assignment to that end-member class. Including LTE as a supplementary data point to anchor the M&S prior did not materially change the posterior M&S mean, but reduced the credible-interval width by approximately 0.5% points, a modest improvement reflecting the limited influence of a single local measurement on a broad literature prior; Because the $\delta^{15}\text{N}-\delta^{18}\text{O}$ ranges of manure and sewage overlap, they were combined into a single manure-and-sewage (M&S) source, so the model resolves three isotopically distinct end-members from 39 summer samples on two isotopes. Isotope fractionation was incorporated through the source-variance and discrimination terms of Eqs. (2)–(5). The model was implemented in SIAR (R v4.1.3; RStudio 2022.02.0-443; MixSIAR formulation of by Markov-chain Monte Carlo (Kiszka et al. 2014; Ren et al. 2022); convergence was confirmed before interpretation, and source proportions are reported as posterior mean $\pm$ SD with 95% credible intervals (Table S4, Text S2).

Results and discussion

Dissolved inorganic nitrogen concentrations

DIN was detected in all river-water samples. River-water TN ranged from 0.24 to 16.11 mg/L, with a mean of 5.02 ± 4.04 mg/L. This mean was lower than the ground-water mean of 6.11 ± 5.01 mg/L, but the difference was not significant ($p > 0.05$). Nitrate was the dominant DIN species in river water, averaging 4.27 ± 3.55 mg/L. Mean NH_4^+ -N and NO_2^- -N were 0.68 ± 0.42 and 0.06 ± 0.04 mg/L, respectively. The same order occurred in groundwater, with means of 5.75 ± 4.82 , 0.36 ± 0.28 , and 0.07 ± 0.05 mg/L, respectively. Nine of the 76 samples (12%) exceeded the WHO/China drinking-water guideline for NO_3^- -N (10 mg/L, equivalent to 50 mg/L as NO_3^-); exceedances were more frequent in groundwater (19%) than in river water (8%) and were concentrated in groundwater near the industrial park and in winter river water at food/grain-processing tributaries (Table S2). The NH_4^+ -N concentration level is significantly ($p < 0.05$) higher than that in large rivers with good water quality, such as the middle reaches of the Yangtze River (0.24 mg/L) and the Pearl River (0.33 mg/L) (Zhang et al. 2011; Xue et al. 2009), and the water quality is roughly the same as that of the main stream of the Yellow River, China, which is greatly affected by local industrial and agricultural activities. Compared with karst river basins abroad, the Lipu River Basin is much better ($p < 0.05$) than the Silver Spring in Florida in USA (1.55 mg/L) and the Coesle River Basin in France (2.34 mg/L). In summary, the water quality demonstrates a higher contamination risk compared to major rivers both in China and internationally, reflecting clear evidence of human impact.

Figure 2 presents the concentrations of nitrogen in river and groundwater samples from the Lipu River Basin. In Fig. 2a, the nitrate nitrogen in the river water during summer (average 2.29 ± 1.02 mg/L) was significantly ($p < 0.05$) lower than that of the other DIN, while there was no significant difference in the four types of ammonia nitrogen and nitrite nitrogen ($p > 0.05$), indicating that nitrate nitrogen dominates the TN and its seasonal variation is relatively large. Figure 2b shows that river-water TN was significantly lower in summer (2.95 ± 1.48 mg/L) than in winter (7.09 ± 6.59 mg/L; $p < 0.05$). Groundwater TN did not differ significantly between seasons. This contrast suggests that the aquifer buffered seasonal variation, whereas river water responded more directly to seasonal hydrological and anthropogenic inputs. This pattern is consistent with rapid conduit flow. Regional faults and folds may provide preferential groundwater pathways (Hu 2024).

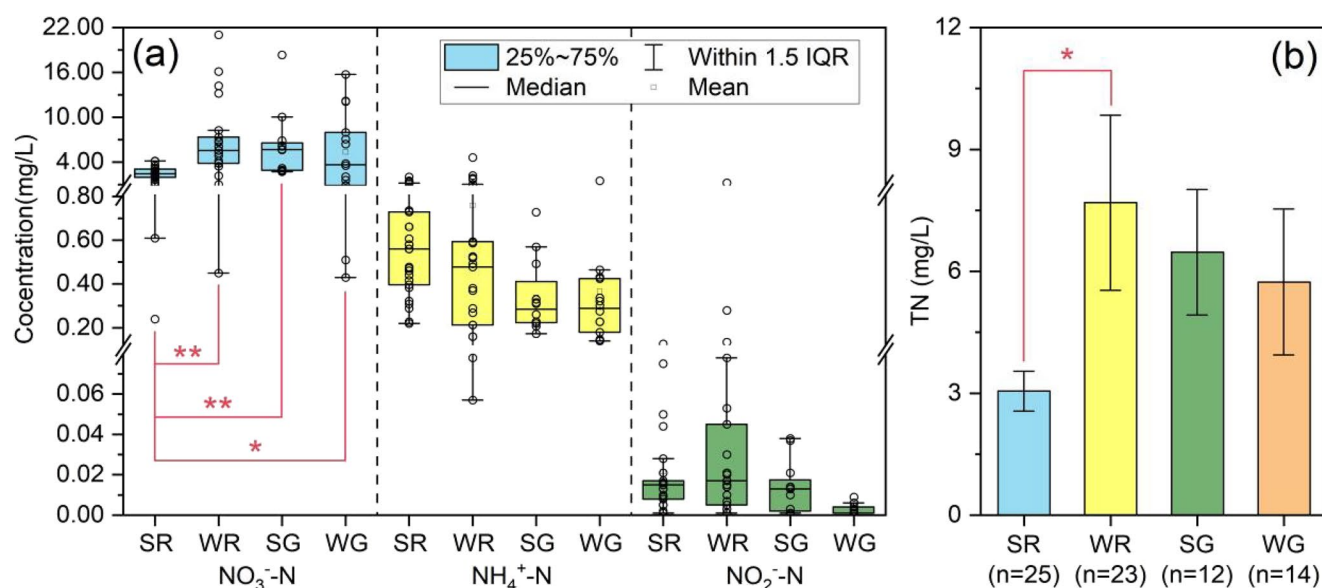


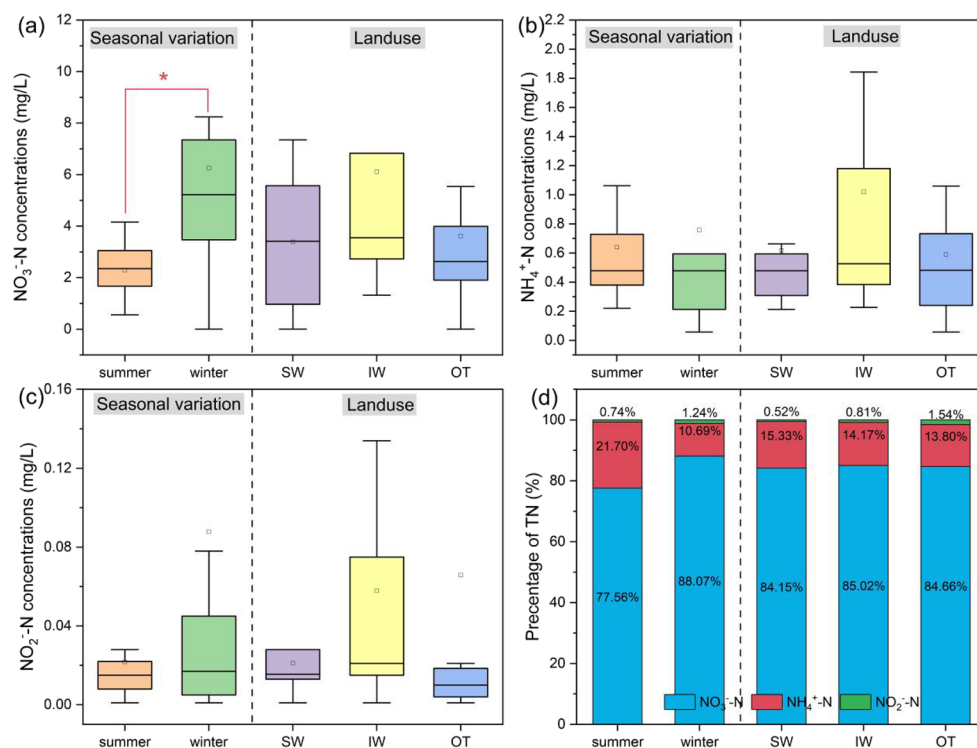
Fig. 2 Concentrations of (a) nitrogen species and (b) total nitrogen (TN) in river water and groundwater during summer and winter seasons. SR, summer river water; WR, winter river water; SG, summer groundwater; WG, winter groundwater. Significance levels: * $p < 0.05$; ** $p < 0.01$

Nitrogen content in river water

During the summer, as shown in Fig. 3, the concentration of NO_3^- -N in the river water ranging from 0.24 to 4.16 (average 2.29 ± 2.01 mg/L), NH_4^+ -N (0.64 ± 0.43 mg/L), NO_2^- -N (0.03 ± 0.01 mg/L), with a relative content of 77.57, 21.68 and 0.75%. NO_3^- -N concentrations ranged from 0.35 to 14.25 mg/L (6.25 ± 5.65 mg/L), NH_4^+ -N (0.76 ± 0.65 mg/L),

NO_2^- -N (0.09 ± 0.05 mg/L), with a relative content of 88.03, 10.70 and 1.27%. Overall, the relative content of NO_3^- -N was more than 77%, indicating that the nitrogen form entering the river was mainly NO_3^- -N, and the sources were mainly nitrified or rapidly nitrified pollution sources, including fertilizer application, livestock breeding, and sewage treatment plant discharge in the study area (Yu Z et al. 2022). The average concentration of NO_3^- -N in

Fig. 3 Spatial and temporal variation in river-water nitrogen. Panels show (a) NO_3^- -N, (b) NH_4^+ -N, and (c) NO_2^- -N by discharge and land-use category, and (d) the proportion of each species in TN. SW, sanitary wastewater; IW, industrial wastewater; OT, other types



winter (6.25 mg/L) was significantly higher ($p < 0.05$) than that in summer (2.29 mg/L), and the fluctuation range was larger. $\text{NH}_4^+\text{-N}$ also increased slightly in winter. The winter increase may reflect stronger external nitrogen inputs, reduced biological uptake, or weaker dilution (Friedrichs et al. 2019).

Near industrial land, mean $\text{NO}_3^-\text{-N}$ was 6.65 ± 5.63 mg/L. Mean $\text{NH}_4^+\text{-N}$ and $\text{NO}_2^-\text{-N}$ were 0.75 ± 0.53 and 0.04 ± 0.03 mg/L, respectively. However, DIN did not differ significantly among industrial, agricultural, and other land-use categories ($p > 0.05$). This lack of separation may reflect mixing among discharges within the river and sewage-collection network (Shoaib et al. 2008).

Nitrogen content in groundwater

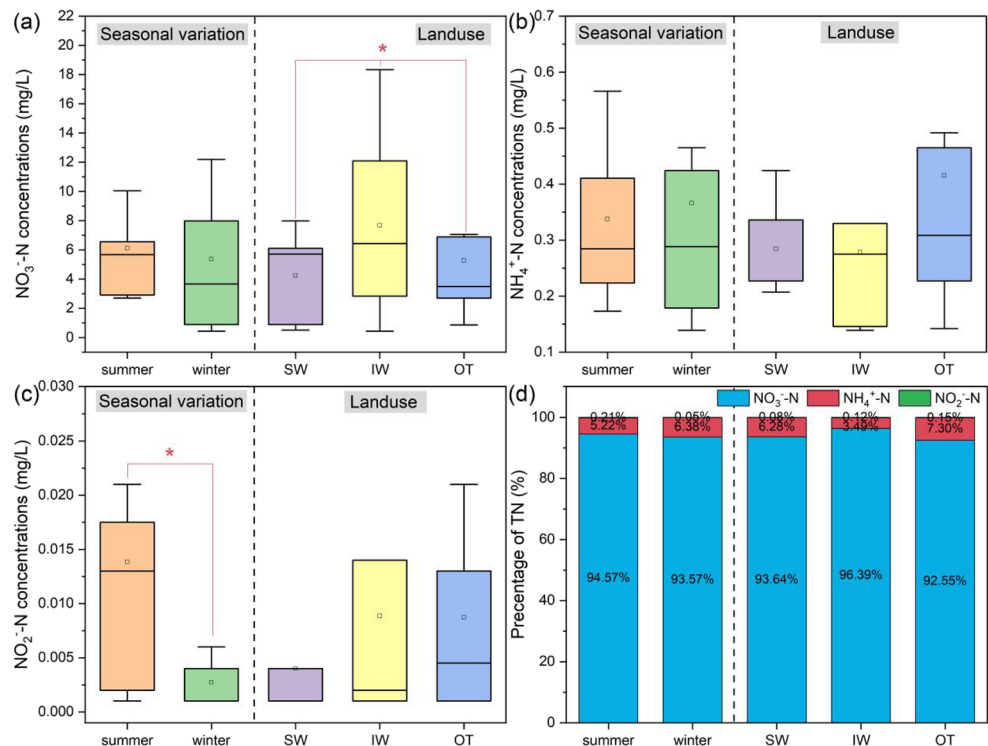
The TN concentration of groundwater in the basin (average 6.12 ± 5.33 mg/L) was higher than that of river water, indicating persistent pollution seepage and accumulation (Fig. 4). In summer groundwater, mean $\text{NO}_3^-\text{-N}$ was 5.98 ± 5.11 mg/L. Mean $\text{NH}_4^+\text{-N}$ and $\text{NO}_2^-\text{-N}$ were 0.34 ± 0.31 and 0.02 ± 0.01 mg/L, respectively, accounting for 5.25% and 0.22% of TN. In winter, mean $\text{NO}_3^-\text{-N}$ and $\text{NH}_4^+\text{-N}$ were 5.36 ± 5.21 and 0.37 ± 0.34 mg/L, respectively. Groundwater $\text{NO}_2^-\text{-N}$ was significantly higher in summer than in winter (Mann-Whitney, $p = 0.009$; medians 0.013 and 0.002 mg N/L, respectively; Table S5). The concentration of $\text{NO}_2^-\text{-N}$ in the summer was significantly ($p < 0.05$) higher than in the winter, with increased rainfall in the summer and

river runoff scouring dissolved $\text{NO}_3^-\text{-N}$ precursors into the vesicular zone for infiltration. In karst basins, rapid infiltration through fissure-dominated flow pathways facilitates pollutant transport to phreatic aquifers during summer, leading to elevated $\text{NO}_2^-\text{-N}$ concentrations (Chu et al. 2021).

Mean groundwater $\text{NO}_3^-\text{-N}$ in the Central High-Tech Industrial Park was 4.53 ± 3.88 mg/L, below the 10 mg N/L guideline; however, individual groundwater samples near the industrial park exceeded the guideline (Table S2). Industrial-wastewater outlets serve chemical, pharmaceutical, and electroplating facilities that use nitrate- and nitrite-containing materials (Huo et al. 2019; Li et al. 2025). Residual nitrogen in the effluent may increase wastewater $\text{NO}_3^-\text{-N}$ concentrations. The nutrients may partially remain in the wastewater, in turn increasing the concentrations of $\text{NO}_3^-\text{-N}$ in the wastewater. The average $\text{NH}_4^+\text{-N}$ in groundwater was 0.41 mg/L, which was slightly lower than the concentration in river water bodies, and the pH of water in karst basins was weakly alkaline (pH 7.5–8.5). Under weakly alkaline conditions, NH_4^+ may be lost through ammonia volatilization before infiltration, which could reduce the amount reaching groundwater (Woodward et al. 2011).

Groundwater $\text{NO}_3^-\text{-N}$ did not differ significantly among industrial, agricultural, and other land-use categories (Kruskal-Wallis, $p = 0.48$; Table S5). The highest individual values nevertheless occurred at wells near the industrial park. Elevated groundwater nitrate may be associated with the hanger-manufacturing and specialty food-processing industries prevalent in Lipu County (Yan et al. 2024). During

Fig. 4 Spatial and temporal variations of nitrogens in groundwater: (a) $\text{NO}_3^-\text{-N}$, (b) $\text{NH}_4^+\text{-N}$, and (c) $\text{NO}_2^-\text{-N}$ concentrations classified by pollutant discharge level and land use type; (d) proportions of nitrogen species relative to TN



wastewater treatment, ammonia oxidation and nitrification convert reduced nitrogen to NO_3^- -N. Organic nitrogen in food-processing wastewater and nitrite oxidation may provide additional nitrate that can subsequently enter groundwater. This finding aligns with Wu's probabilistic assessments (Wu et al. 2025). The absence of significant nitrate differences among river water categories does not imply that their discharge sources are identical. Instead, it may indicate stronger plume mixing in the river than in the aquifer.

Spatial distribution

Nitrogen content in river water

During the observation period, river-water TN averaged 4.95 ± 4.36 mg/L. Using the WHO and Chinese nitrate guideline, 8% of river sites exceeded the threshold. Mean nitrogen concentrations were relatively high in the Puling, Dumo, Datang, and Limu tributaries. Concentrations in the lower Maling River were 2.2 mg/L higher than upstream values, possibly reflecting inputs from the nearby grain- and food-processing industrial park. The concentrations of various forms of nitrogen in the mainstream are NH_4^+ -N (0.55 ± 0.38 mg/L), NO_3^- -N (2.07 ± 1.78 mg/L), and NO_2^- -N (0.02 ± 0.01 mg/L). In the tributaries, they are respectively NH_4^+ -N (2.21 ± 1.87 mg/L), NO_3^- -N (2.52 ± 2.38 mg/L), and NO_2^- -N (0.01 ± 0.01 mg/L). Along the Lipu mainstream, nitrogen concentrations were lower in the middle reach and higher in the upper and lower reaches. The middle reach passes through urban Lipu, whereas site DT2 receives inputs from several industrial sectors, including metal, oil, rice, and flour processing. Strong karst development in the tributaries leads to low DO (< 2 mg/L), and an anoxic environment severely inhibits nitrification (Diab et al. 1993), resulting in the accumulation of NH_4^+ -N in the tributaries. Mean TN was higher in the tributaries (5.11 mg/L) than in the mainstream (2.64 mg/L). This difference was driven mainly by elevated NH_4^+ -N at a few tributary sites, whereas the NO_3^- -N difference was not statistically significant (Mann-Whitney, $p=0.11$; Table S5), and the flow rates in tributaries were mostly over $3.5 \text{ m}^3/\text{s}$, which is more than 30% of the flow rate of the main stream. The pollution load is relatively high, indicating that tributaries such as the Pulu and Dumo Rivers are the core sources of N pollution in the basin (Fakouri et al. 2022). The river water monitoring points were mainly driven by OT and SW.

Points L5, DM1, and DM2 were affected by nearby karst skylights, with higher DO (5 mg/L) and NO_3^- -N concentrations of 3.05, 4.16, and 2.93 mg/L, respectively. Near PL2, L6, and DT2, there were large karst springs

with NO_3^- -N concentrations of 3.11, 2.97 and 3.05 mg/L respectively, all higher than the average NO_3^- -N concentration (2.88 mg/L) in the basin, confirming the earlier conclusion of groundwater recharge to the river water. Considering the regional monitoring points were pure carbonate rock hydrogeological conditions, the retention time of pollutants around the large springs and skylights was potentially shortened (Meng et al. 2021) and the natural degradation rate was relatively low.

Nitrate-nitrogen content in groundwater

Groundwater TN averaged 5.743 mg/L, and 18.83% of groundwater sites exceeded the WHO and Chinese nitrate guideline. TN at sites U10 and U11, near the metalworking industrial park, was two to three times the regional mean (Fig. 5c, d). The area near the industrial-wastewater outlet was therefore a nitrate hotspot and may have contributed to elevated concentrations in surrounding wells. The upstream agricultural area has become a significant source of groundwater NO_3^- -N pollution, and the fundamental driving factor is extensive application of organic fertilizer in the Lipu Taro planting model and the subsequent nitrification process. The concentrations of NO_3^- -N in the U1–U3 region were 1.65–3.57 mg/L higher than the average concentration of groundwater in the region. Organic fertilizers can account for 30%–50% of the cultivation of Lipu taro in the region. Organic nitrogen is converted into NH_4^+ -N through microbial mineralization and then into NO_3^- -N by nitrifying bacteria. If the application is excessive or uncomposted, the rate of nitrate formation exceeds the absorption capacity of the crop (Cui et al. 2023 b), resulting in accumulation of NO_3^- -N. U4 and U5 were affected by nearby karst skylights, with NO_3^- -N concentrations of 2.87 and 3.2 mg/L, respectively. There are large karst springs near U10, U11, and U12, with NO_3^- -N concentrations of 6.1, 18.3, and 5.6 mg/L, respectively. The monitoring data near the karst spring showed no significant differences from the other points in the basin; however, the monitoring points near the karst skylight showed significant differences from those of the other points ($P < 0.05$). Sites U10–U12 sampled bedrock-fissure groundwater with limited precipitation recharge, short subsurface flow paths, and low discharge (Yi et al. 2023). Spring flow ranged from 0.014 to 0.325 L/s. Mean NO_3^- -N across U10–U12 was approximately 10 mg/L, possibly reflecting inputs from the nearby new-materials industrial park.

NO_3^- -N and dissolved oxygen varied among sites near karst springs, skylights, and porous-aquifer wells. These spatial patterns are consistent with heterogeneous conduit and matrix flow paths, but the available groups do not

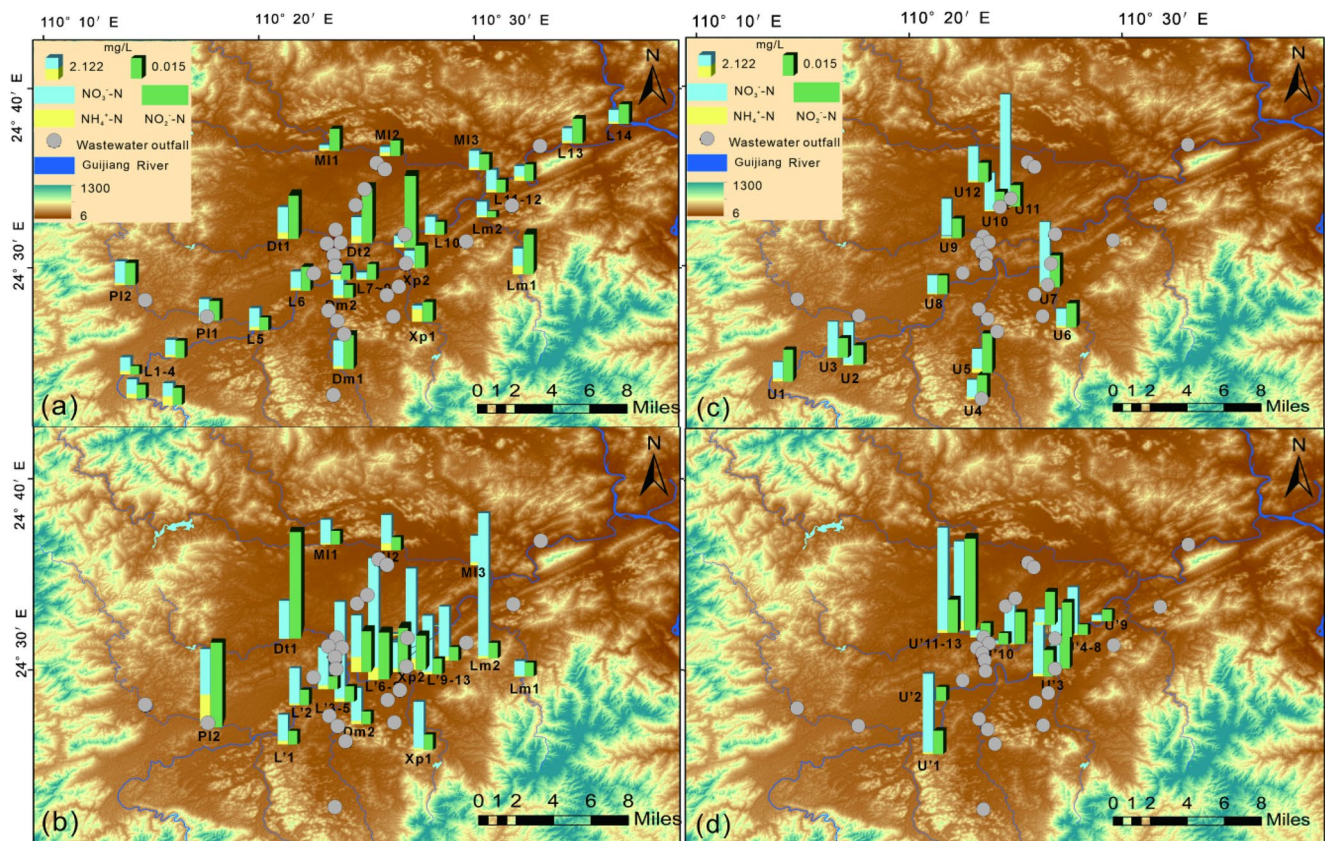


Fig. 5 Spatial distribution of nitrogen in river water and groundwater during summer and winter. (a) Summer river water; (b) winter river water; (c) summer groundwater; (d) winter groundwater

support a single statistically significant conduit-versus-matrix contrast. The U1-U3 wells represent a Quaternary porous aquifer, whereas several other sites are associated with karst springs or skylights; this hydrogeological distinction is therefore treated as a process interpretation.

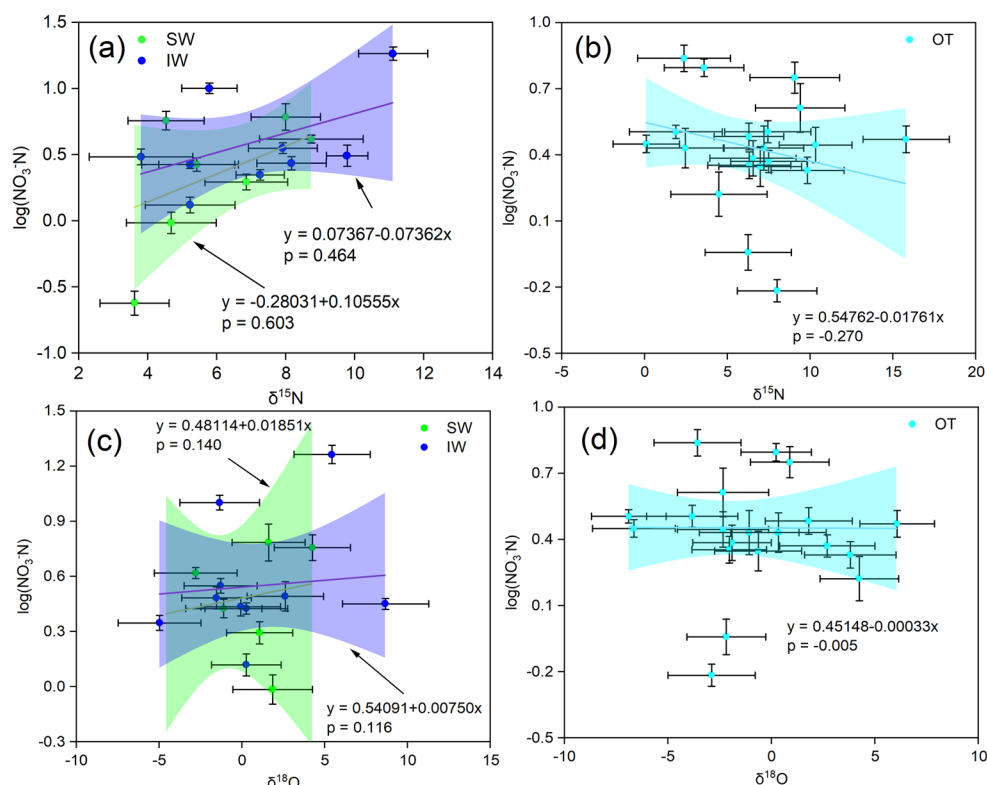
Seasonal variation

In summer, mean TN was lower in river water (2.96 ± 2.36 mg/L) than in groundwater (6.74 ± 5.36 mg/L). In winter, the pattern reversed: river-water TN averaged 7.12 ± 6.65 mg/L, compared with 5.77 ± 4.86 mg/L in groundwater (Figs. 5 and 6). River-water NO₃⁻-N, NH₄⁺-N, and NO₂⁻-N were higher from October to March than from July to September, likely because summer rainfall and runoff diluted river concentrations. During the winter, there is less precipitation and water flow slows down, and NO₃⁻-N accumulation in the basin leads to higher concentration. Sewage outlets within the basin also affect the distribution of nitrogen; however, the dilution and accumulation effects caused by seasonal changes are more significant. Notably, there was a significant difference in NO₂⁻-N between the

wet and dry seasons, with concentrations in the summer being approximately three times those in the winter. The unique hydrological and water quality conditions in the study area during the summer (high load input, turbidity, and anoxic microenvironment) disrupted nitrogen transformation (Zhao et al. 2020) and inhibited the nitrite consumption pathway (Fig. 5b). In groundwater, the high value areas of the DIN are often concentrated at the IW monitoring point (Fig. 5cd), which reflects both the continuous influence of industrial pollution sources and the fissure-pipe type flow characteristics of karst basins. Once pollutants enter a karst aquifer through infiltration, they migrate to relatively connected channels, forming relatively stable high-value areas.

Between summer and winter, the contribution of NO₃⁻-N to total DIN in river water declined from 88.07% (winter) to 77.56% (summer). In summer, dilution by high rainfall reduces absolute concentrations while the dominant signal shifts towards non-point (agricultural) sources, partially masking point-source contributions. In winter, lower discharge concentrates the point-source signal; elevated NO₃⁻-N at DT1, LM2 and L13, combined with elevated Cl⁻ and EC, indicates direct industrial and sanitary inputs.

Fig. 6 Relationships of log-transformed NO_3^- concentrations with $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in the Lipu River Basin. Shading indicates the confidence interval around the fitted mean. Panels (a, b) show relationships with $\delta^{15}\text{N}$, and panels (c, d) show relationships with $\delta^{18}\text{O}$



Hydrochemical and isotopic constraints on NO_3^- sources and transformation

Two independent diagnostics are used here to identify the processes controlling NO_3^- rather than its sources: the nitrogen enrichment slope (NES, Eq. 1), which relates $\log C(\text{NO}_3^-)$ to $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$, and the conservative tracers Cl^- and EC, whose covariation with NO_3^- distinguishes source mixing from in-situ removal. Together they constrain the extent of denitrification and provide the process framework within which the source apportionment in Section “SIAR-based nitrate source apportionment” is interpreted.

Nitrogen enrichment slope: $\log C(\text{NO}_3^-)$ versus $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$

At the SW and IW outlets the NES was positive, $\log C(\text{NO}_3^-)$ increasing with $\delta^{15}\text{N}$ (Fig. 6a, b). This is the signature expected where manure- and sewage-derived NO_3^- accumulates along the flow path, because these sources carry high $\delta^{15}\text{N}$, generally +10 to +25‰ (Serna et al. 2010). At the OT outlets the slope was negative. The most probable control is the low $\delta^{15}\text{N}$ of synthetic fertilisers (urea, ammonium salts and nitrate fertilisers), which is close to the atmospheric value of 0‰ and generally ranges from −6 to +6‰: where fertiliser is applied in excess of plant uptake and soil fixation, runoff and leaching deliver high NO_3^- concentrations at low $\delta^{15}\text{N}$ (Šturm et al. 2011). The contrast in the sign of

the NES between SW/IW and OT therefore separates the two dominant source families independently of, and prior to, the mixing model. The fitted relationships and their coefficients of determination are shown in Fig. 6.

The regressions against $\delta^{18}\text{O}$ were likewise positive at SW and IW (Fig. 6c, d). Because nitrification incorporates oxygen from ambient water, NO_3^- produced in situ inherits a $\delta^{18}\text{O}$ close to that of the local water, so a positive $\log C(\text{NO}_3^-)$ - $\delta^{18}\text{O}$ relationship indicates that NO_3^- at these outlets is largely nitrification-derived while the contributing sources (domestic, industrial and soil mineralisation) retain distinguishable $\delta^{18}\text{O}$ signatures (Buchwald et al. 2012). The weaker relationship at OT is consistent with a larger, isotopically distinct fertiliser contribution. Critically, neither regression shows the joint enrichment of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ with falling NO_3^- that would indicate denitrification; that possibility is evaluated quantitatively in Section “Extent of denitrification”.

Conservative tracers (Cl^- and EC): mixing versus in-situ removal

Cl^- behaves conservatively in these waters and EC integrates total mineralisation, so their covariation with NO_3^- discriminates between the two processes that can control NO_3^- concentration (Xu et al. 2016). If NO_3^- rises together with Cl^- and with EC, the concentration change is governed by source mixing and conservative transport; if

denitrification is active, NO_3^- falls while Cl^- is unchanged, so that the $\text{NO}_3^-/\text{Cl}^-$ ratio declines as $\delta^{15}\text{N}$ rises. The two diagnostics are applied separately to river water and to groundwater, because Section “[Spatial distribution](#)” showed that the two compartments differ in flow path and residence time.

The group-specific regressions in Fig. 7 showed that the NO_3^- -EC relationship was strongest for SW ($R^2 = 0.593$), followed by IW ($R^2 = 0.464$), while OT showed only a weak relationship ($R^2 = 0.032$). The NO_3^- - Cl^- relationship was also strongest for SW ($R^2 = 0.483$); the corresponding values for OT and IW were 0.216 and 0.206, respectively. The SW and IW patterns are consistent with co-discharge and conservative mixing, whereas the weaker OT relationships indicate more heterogeneous inputs. These regressions do not by themselves quantify denitrification, which is evaluated using the isotope evidence in Section “[Extent of denitrification](#)” (Fig. 7; Table S5). In river water, NO_3^- was positively correlated with both EC and Cl^- at all discharge types (Fig. 7a, b), and the relationship was strongest at the industrial outlets ($R^2 = 0.596$), where chloride-rich effluent is expected and where elevated wastewater temperature further increases ionic mobility and hence EC (Zhang et al. 2020). Read together with the winter point-source signal described in Section “[Seasonal variation](#)”, the positive NO_3^- - Cl^- relationship (Table S5) indicates that river-water NO_3^- is controlled by conservative mixing of discharge sources rather than by in-stream removal. In groundwater, by contrast, the NO_3^- - Cl^- relationship was not significant. This decoupling cannot be produced by mixing alone and is attributed to vertical karst heterogeneity: where matrix-dominated, hypoxic to anoxic conditions develop (Lorette et al. 2022), NO_3^- is removed locally while Cl^- is conserved. The conduit-matrix contrast in dissolved oxygen reported in Section “[Nitrate-nitrogen content in groundwater](#)” thus has a measurable geochemical expression, whose magnitude is bounded below.

Extent of denitrification

The $\delta^{18}\text{O}$ - NO_3^- values of all samples fall within the range produced by nitrification, confirming that nitrification, rather than direct input of oxidised nitrate, establishes the NO_3^- pool in both compartments. Where denitrification does occur, nitrogen and oxygen isotopes fractionate according to a Rayleigh model, so $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ both increase as NO_3^- decreases, with an enrichment ratio $\varepsilon_{\text{N}}/\varepsilon_{\text{O}}$ of about 1.3–2.1. Most river-water samples plotted to the left of the $\varepsilon_{\text{N}}/\varepsilon_{\text{O}} = 2.1$ line and showed no coupled increase in $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$, that is, no resolvable denitrification. In groundwater, $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ were inversely related to NO_3^- -N with an enrichment ratio of about 1.38, which does indicate denitrification (Plata et al. 2025); the δ values nevertheless remained low and the samples were only suboxic ($\text{DO} < 2 \text{ mg/L}$), so the extent of removal was limited. This bound is consistent with the loss of the NO_3^- - Cl^- relationship in groundwater (Section “[Conservative tracers \(\$\text{Cl}^-\$ and EC\): mixing versus in-situ removal](#)”) and means that the proportions derived in Section “[SIAR-based nitrate source apportionment](#)” can be treated as near-conservative mixtures, with the caveat that the groundwater M&S contribution may be marginally underestimated.

SIAR-based nitrate source apportionment

The SIAR model estimated source contributions to NO_3^- -N in river water and groundwater (Fig. 8). Manure and sewage were combined as M&S because their isotope ranges overlapped. Posterior estimates are reported as means $\pm$ SD with 95% credible intervals (Text S2). M&S was the largest modeled source in river water, followed by soil nitrogen. The soil contribution may reflect runoff that transported organic nitrogen from residential soils during heavy summer rainfall (Cui et al. 2020), and heavy rainfall during the summer washed the homestead soil, carrying high amounts of organic nitrogen into the river water. After entering river water, some organic nitrogen may be converted to nitrate

Fig. 7 (a) Relationship between nitrate (NO_3^-) and electrical conductivity. (b) Relationship between NO_3^- and Cl^-

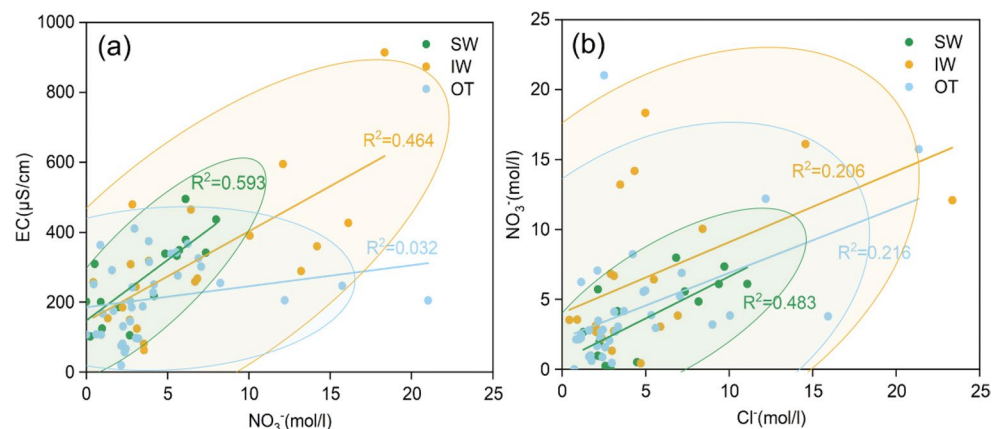
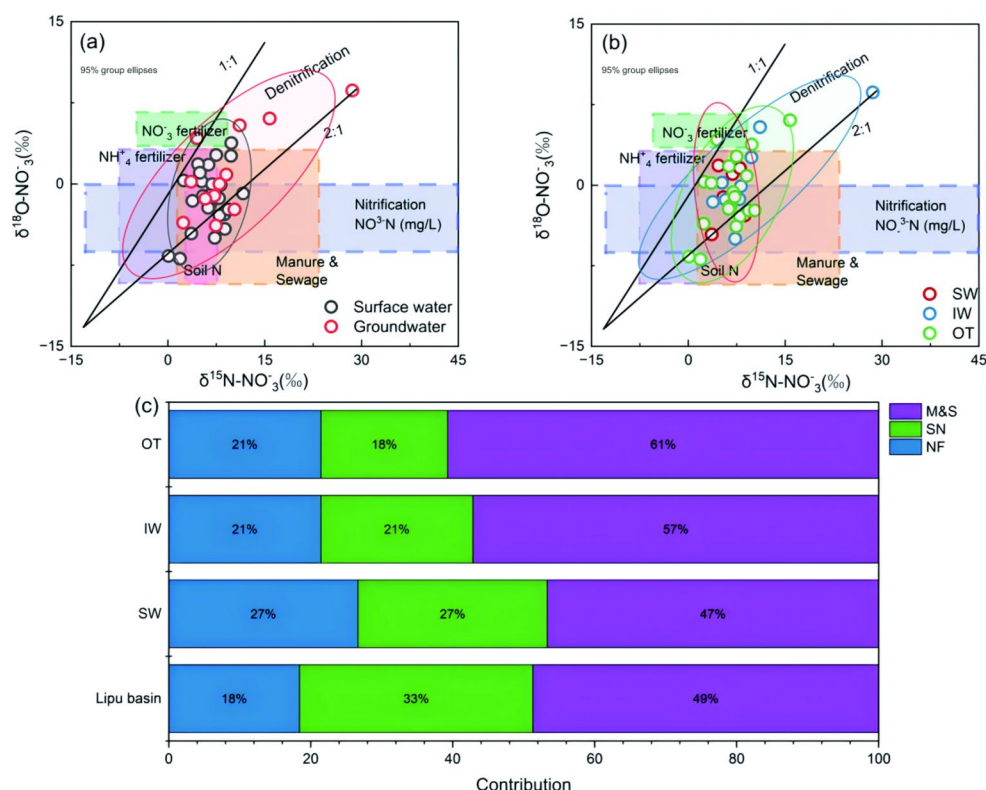


Fig. 8 Distribution of $\delta^{15}\text{N}-\text{NO}_3^-$ and $\delta^{18}\text{O}-\text{NO}_3^-$ values and modeled nitrate-source proportions. (a) River water and groundwater; (b) sanitary wastewater (SW), industrial wastewater (IW), and other types (OT); (c) SIAR proportions for the three modeled source categories by discharge category and for the whole basin. Shaded outlines in panels (a) and (b) show 95% group ellipses. Rounded values in panel (c) are listed in Table S7



through nitrification rather than transported directly to groundwater (Xiong et al. 2023). Soil nitrogen was nevertheless the second-largest modeled groundwater source, possibly through river recharge or direct soil leaching. M&S was the largest modeled groundwater source. The proposed role of landfill flushing and infiltration should be stated only if supported by site-specific evidence; otherwise, present it as a hypothesis rather than an explanation (Vadas et al. 2008). Isotope-based apportionment in karst systems is subject to uncertainties arising from overlapping end-member signatures, conduit–matrix mixing, and variable nitrification/denitrification. The reported proportions therefore represent the summer basin-scale NO_3^- budget, with associated uncertainty reflected by the credible intervals (Text S2).

Quantitatively, the posterior estimates confirmed M&S as the dominant NO_3^- source. For the basin as a whole, M&S contributed $56.7 \pm 8.2\%$, followed by soil nitrogen ($22.5 \pm 6.5\%$), nitrogen fertilizer ($15.3 \pm 5.0\%$) and atmospheric deposition ($5.5 \pm 2.5\%$). The M&S contribution was higher in groundwater ($61.5 \pm 8.5\%$) than in river water ($50.2 \pm 8.0\%$), whereas soil nitrogen was correspondingly higher in river water ($27.0 \pm 6.8\%$) than in groundwater ($21.0 \pm 6.2\%$); NF and AD were minor in both (Table S7). The very low AD contribution is consistent with the low

$\delta^{18}\text{O}-\text{NO}_3^-$ values of the samples, which lie well below the atmospheric end-member.

The $\delta^{18}\text{O}-\text{NO}_3^-$ values of the water samples in the study area were all within the typical $\delta^{18}\text{O}-\text{NO}_3^-$ range produced by nitrification (Li et al. 2019), indicating that nitrification is a key process for establishment of the source of NO_3^- in river water and groundwater. In groundwater dominated by nitrification, the higher the dissolved oxygen content, the higher the $\delta^{15}\text{N}-\text{NO}_3^-$ value. The nitrogen and oxygen isotope fractionation produced during denitrification followed Rayleigh fractionation, and the $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ values of the remaining NO_3^- satisfied the Rayleigh formula, showing that as NO_3^- concentration decreased, both $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ values increased and were linearly correlated. In addition, the slope of the enrichment coefficient ratio ($\epsilon_{\text{N}}/\epsilon_{\text{O}}$) varied between 1.3 and 2.1. Most of the data at the monitoring points were concentrated in the left zone of $\epsilon_{\text{N}}/\epsilon_{\text{O}} = 2.1$, where the NO_3^- -N concentration and the values of nitrogen and oxygen isotopes were within a certain range, indicating that the changes in the water body were not significant and there was no obvious denitrification. Groundwater $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ values were inversely related to NO_3^- -N concentrations, with an enrichment-slope ratio of approximately 1.38. This pattern was consistent with some denitrification (Yuan et al. 2020), although the

generally low isotope values suggested that the process was limited. The interpretation of $\text{DO} < 2 \text{ mg/L}$ as evidence of weak denitrification should be justified, because suboxic conditions can also favor denitrification. The $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ values in groundwater were slightly higher than those in river water, suggesting that groundwater was more strongly influenced by fecal matter and domestic sewage inputs. This is likely attributable to the widespread use of dry toilet systems in rural areas and the application of manure to agricultural land (Wang et al. 2021).

Compared with other karst and plateau systems in Southwest China, the M&S contribution in the Lipu River basin (56.7%) is notably higher. In the Babu karst aquifer, manure and sewage, nitrogen fertilizer, and soil nitrogen contributed 37%–38%, 31%–32%, and 31%, respectively (Ren et al. 2022). In the Erhai Basin, manure and sewage contributed 49.6%, while soil nitrogen contributed 37.2%. The higher M&S contribution in the Lipu Basin may reflect dense rural settlement, dry-toilet use, manure application, and local industrial discharge of dissolved-organic-nitrogen-rich wastewater (Table S8). These findings suggest that M&S is a major nitrogen source in karst groundwater in Southwest China, whereas the Lipu River Basin is also affected by local industrial activities.

The $\delta^{18}\text{O}\text{-NO}_3^-$ of agricultural area was higher than that of other areas, suggesting that major nitrogen biogeochemical processes have changed across different pollution types. Many studies have found that pollution type and water body characteristics largely influence the NO_3^- isotopic characteristics (Choi et al. 2020; Yu et al. 2022a, b). The $\delta^{15}\text{N}\text{-NO}_3^-$ values observed in IA were lower than those observed in FA, mainly because SN and NF are the main sources of NO_3^- (Kim et al. 1996). Isotope enrichment in river water may reflect increased fertilizer use, enhanced manure volatilization, and evaporation of waste-derived leachate during warm periods. However, $\delta^{18}\text{O}\text{-NO}_3^-$ may respond more strongly to microbial nitrification than to source, land use, or season. The higher groundwater $\delta^{15}\text{N}\text{-NO}_3^-$ values were consistent with M&S inputs from densely populated areas and waste leachate.

Compared with other karst and plateau systems in Southwest China, the M&S contribution in the Lipu River basin ($56.7 \pm 8.2\%$) is notably higher than that reported for the Babu karst aquifer, Guizhou (37–38%), and the Erhai Basin on the Yunnan–Guizhou Plateau (49.6%), where nitrogen fertilizer and/or soil nitrogen remain co-dominant. The greater dominance of M&S in the Lipu basin reflects its specific land-use setting—dense rural settlements with widespread dry-toilet systems and manure application, combined with hanger-manufacturing and taro food-processing

industries that discharge dissolved DON-rich wastewater—rather than the fertilizer or soil-dominated signatures typical of more strictly agricultural karst catchments. Table S8 summarises this cross-study comparison.

Conclusion

This study combined dual nitrate isotopes, hydrochemistry, and SIAR modeling to evaluate nitrate sources and transformations in a coupled karst river-groundwater system. Groundwater TN exceeded river-water TN in summer, whereas river-water TN was higher in winter. Groundwater near the metalworking and new-materials park and tributaries near the grain and food-processing park contained localized NO_3^- -N hotspots. In Fig. 7, the strongest NO_3^- -EC relationship occurred at SW outlets ($R^2 = 0.593$), followed by IW outlets ($R^2 = 0.464$); the NO_3^- -Cl⁻ relationship was also strongest for SW ($R^2 = 0.483$). Measured $\delta^{18}\text{O}\text{-NO}_3^-$ values fell within the nitrification-derived range, and an $\epsilon\text{N}/\epsilon\text{O}$ slope of approximately 1.38 suggested limited groundwater denitrification. The three-source SIAR model identified M&S as the largest modeled summer source, contributing 49% at the basin scale. Because nitrate isotopes were measured only in summer, these source proportions should not be interpreted as seasonal estimates. Management should prioritize manure and sewage control in karst-sensitive zones, localized industrial groundwater hotspots, and winter monitoring of tributaries affected by food-processing activities.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s12665-026-13136-4>.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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